Ashmit Mukherjee

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Education

New York University Abu Dhabi *Expected May 2027*
B.S. in Computer Science; minor in Economics
Study away: NYU Paris (Spring 2025); NYU New York (Fall 2025)
Selected coursework: Social Networks; Natural Language Processing; Principles of Data Science I; Statistics for the Social and Behavioral Sciences; Algorithms

Research Interests

Computational social science; human–AI interaction; experimental methods; empirical machine learning.

Research Experience

Research Assistant, Human–AI Collaboration *Feb 2026 – present*
NYU Abu Dhabi | With Ben Rosche and Hanan Salam

- Built the software for a faculty-led study of AI advice and cooperation: Python/FastAPI game mechanics, Empirica participant and researcher workflows, AI integration, session recovery, and data exports.
- Enforced what each adviser can see, kept recommendations separate from decisions, and made interrupted sessions resume from the recorded decision rather than applying it twice.
- Built checks that reconstruct outcomes from recorded actions instead of trusting the engine's totals. The platform is in development, with no human-study results yet.

Research Assistant, eBRAIN Lab *Feb 2026 – present*
NYU Abu Dhabi | With Muhammad Shafique and Minghao Shao

- Extended protein-model evaluation to signal-peptide classification and cleavage-site prediction, building the ESM-2 task adapters, model heads, and metrics.
- Ran LoRA, full fine-tuning, and biological-prior comparisons on the university HPC cluster across 150M, 650M, and 3B models; investigated initialization, feature injection, and efficiency as distinct experimental choices.
- In a three-seed 650M follow-up, measured 86.08% mean exact cleavage-site match for BLOSUM62 feature injection versus 83.64% for vanilla LoRA on 1,231 signal-peptide-positive examples. Full fine-tuning led this metric at 150M and 3B.
- Reassessed two promising single-seed settings across three seeds: removing weight decay and reducing the initial feature-injection scale both fell below the original BLOSUM62 setting in mean exact cleavage-site match.

Research Manuscript

D. S. Abubaker, M. Faheem, **A. S. Mukherjee**, S. Othman, S. Sifar, S. Sthapit, Y. Wang, M. Shao, and M. Shafique.
Bio-LoRA: Biological Priors as Inductive Bias for Low-Rank Adaptation.
Unpublished manuscript, 2026. Co-first author. ashmitmukherjee.com/work/protein-language-models

Independent Research

Synthetic Pilot Lab, Independent Capstone *Oct 2026 – present*
In development; offline diagnostics on synthetic records, with no human participants

- Built paired correct and faulty game engines, scripted decisions to replay through them, and an independent evaluator that reconstructs expected states and payoffs from submitted actions.
- In a scripted synthetic benchmark, an engine that assigned contributions to the wrong players passed checks on group totals; recalculating each player's payoff exposed it.

Selected Projects

RoomWise, Chair-Fit Checker (github.com/Ansester/RoomWise) *Sep – Nov 2026*

- Co-developed with Aadil Chasmawala an iPhone and desktop prototype that checks a chair's footprint against a captured room, using SwiftUI, RoomPlan, ARKit, and shared Kotlin Multiplatform geometry.
- A hosted language model only interprets the request. Geometry code decides the fit, and missing or conflicting listing facts stay unknown.
- 67 automated tests pass in public CI; not yet measured in a real room or tested with users.

FieldLens, Evidence-Linked Incident Review *Oct 2026*

- Built a local-model incident-review prototype using fictional records, with retrieval limits, exact source references, and a deterministic fallback for inference failures.
- Set validity, coverage, and relevance criteria and compared three answer methods on matched synthetic cases; some answers cited sources correctly and still missed what was asked.

Salary Regression, Course Project *Fall 2025*

- Repaired a leaky train/test split by deduplicating postings and grouping the split, then re-ran the comparison. On 92 held-out postings, XGBoost reached \$23.8k mean absolute error against \$26.9k for a mean baseline (R^2 0.19). This supersedes an earlier R^2 of 0.57, which is not directly comparable.

Hinglish Named-Entity Recognition *Fall 2025*

- Contributed training pipelines, evaluation tooling, and error analysis in a four-person project using mBERT and XLM-RoBERTa on COMI-LINGUA.

CAMP, Campus Asset Management Platform *Fall 2025*

- Built staff-workflow REST endpoints, alerts, and dashboard statistics in a five-person React, Express, and MongoDB project; configured backend integration testing and client API utilities.
- Served as Product Owner and Scrum Master in rotating team sprints.

Applied Experience

Machine Learning Engineer Intern, Zeek *Jun – Aug 2025*

- Built discovery, ROI prediction, explainability, feedback, and campaign components of a marketing ML system in a four-person team, using XGBoost, clustering, SHAP, and T5/GPT.
- Developed and tested on synthetic data only; no real-customer results.

Market Research Consultant, Global Media Alliance Broadcasting *Jun 2024*

- Conducted audience research and designed surveys; presented findings to senior leadership.

Service

First Year Program Facilitator, NYU Abu Dhabi *Aug – Dec 2024*
Facilitated workshops, group activities, and weekly check-ins for more than 20 first-year students.
Front of House Staff, NYU Abu Dhabi Arts Center *Jan – May 2024*
Supported guest check-in, ticketing, and event operations for audiences exceeding 500 attendees.

Technical Skills

Programming and data: Python, JavaScript, C++, Kotlin, Swift, pandas, NumPy, SciPy
Machine learning: PyTorch, Transformers, scikit-learn, XGBoost, SHAP, LoRA
Systems: FastAPI, Empirica, Slurm, pytest, Git, GitHub Actions, SwiftUI, ARKit, RoomPlan